

A Secure Routing Protocol for Wireless Sensor Networks

Jaydip Sen and Arijit Ukil

Innovation Lab, Tata Consultancy Services Ltd., Bengal Intelligent Park,
Salt Lake Electronics Complex, Kolkata, India
{Jaydip.Sen,Arijit.Ukil}@tcs.com

Abstract. Wireless sensor networking has been a subject of extensive research efforts in the recent years, and has been well recognized as a ubiquitous and general approach for some emerging applications such as a real-time traffic monitoring, ecosystem and battlefield surveillance. Since these networks deal with sensitive data, it is imperative that they are made secure against various types of attacks such as node capture, physical tampering, eavesdropping, denial of service etc. This paper presents a secure routing mechanism for wireless sensor networks. The protocol is resilient in the presence of malicious nodes that may launch selective packet dropping attack on the routing path. The scheme employs single-path routing, and therefore, it is energy-efficient. While the packets are forwarded towards the base station, if any node fails to forward a packet, it is isolated immediately. Packets are then routed around the node. Simulation conducted on the scheme clearly demonstrates that it is more efficient than some of the existing similar schemes.

Keywords: Wireless sensor networks, routing, malicious packet dropping, neighborhood watch, energy-efficiency, reliability.

1 Introduction

Recent convergence of technological and application trends has resulted in an exceptional level of interest in wireless ad hoc networks, and in particular, wireless sensor networks (WSNs). The push was provided by rapid progress in computation and communication technology as well as the emerging field of low cost, reliable, MEMS-based sensors, while the pull was provided by numerous applications that can be summarized under the umbrella of computational worlds, where the physical world can be observed and influenced through the Internet and WSN infrastructures. Typically, WSNs contain hundreds or thousands of sensor nodes that have the ability to communicate among them and also to a base station (BS) [1]. While the sensor nodes have limited sensing region, processing power and energy, networking a large number of such nodes makes a robust, reliable and accurate sensor network covering a wider region. The WSNs are envisioned to play an important role in a wide variety of applications ranging from critical military surveillance to forest fire monitoring and the building security monitoring. Since these networks are mostly deployed in harsh and often hostile operating environments, security and reliability are two important aspects for their operations. Design of a robust and secure routing protocol for WSNs is

a particularly challenging problem due to limited computational resources of the nodes and unreliable wireless links. In addition, data transmission through these networks is prone to many types of failures and attacks such as node capture, physical tampering, eavesdropping, denial of service, etc [2, 3, 4]. Some of the previous research efforts in secure routing for WSNs have focused on the design of single-path routing using low-rate and periodic flooding of events to avoid failed nodes [5]. A fault-tolerant cluster-based routing scheme has been proposed in [6]. In [24], a multi-path routing protocol has been presented that uses three paths from the source node to the sink to provide robustness.

To address the problem of security and efficiency in routing in WSNs, this paper proposes a scheme that reliably identifies compromised (or faulty) nodes and utilizes a routing path that avoids these nodes. Essentially, it utilizes a single-path routing concept and thereby saves energy-consumption. If a malicious node is detected on the next-hop on the routing path, the node is efficiently bypassed and the packets are routed around the node to the base station still in a single-path. The proposed protocol is a modification of the routing scheme proposed in [7]. However, it is more energy-efficient and less delay-inducing as observed from the simulation results.

The rest of the paper is organized as follows. Section 2 discusses some related work in the area of secure routing in WSNs. Section 3 presents the WSN model and possible types of packet dropping attacks on these networks. It also gives a brief introduction to the LEAP protocol for key management used in the proposed scheme. Section 4 gives a detailed description of the proposed protocol. Section 5 presents the simulation results and Section 6 concludes the paper.

2 Related Work

A considerable amount of research effort has been put in design of secure routing protocols for WSNs. While some of these works have focused on defense against outsider attacks on the key management protocols being used in the network [8, 9, 10], the others have attempted to secure node-to-node communication [10, 11]. However, all these schemes fail even if a single node is compromised.

An insider attack on a WSN is even more dangerous, since an attacker in this case can launch various types of attacks such as dropping of legitimate packets, injecting of bogus packets and sensing reports, advertising false routing messages, eavesdropping on the network communication etc. Parno et al have proposed a distributed detection mechanism [12]. Newsome et al [13] have presented some techniques for preventing an adversary from launching Sybil attack by arbitrarily creating new identities.

Deng et al have proposed an *intrusion tolerant routing protocol* in wireless sensor networks (INSENS) that adopts a routing-based approach to security in WSNs [14]. It constructs routing tables in each node, bypassing malicious nodes in the network. The protocol cannot totally prevent attack on nodes, but it minimizes the damage caused to the network due to an attack. The scheme has reduced computation, communication, storage, and bandwidth requirements at the sensor nodes at the cost of greater computation and communication overhead at the base station. Tanachaiwiwat et al have proposed a novel secure routing protocol- *trust routing for location-aware*